



diffpy.cmi tutorial

ADD 2026

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Agenda

- What is diffpy.cmi?
- Setting up a modeling problem
- Installing diffpy.cmi
- Examples
 - Fitting noisy data
 - Fitting a SrFe_2As_2 PDF
 - Fitting SrFe_2As_2 temp series
 - ...

What is diffpy.cmi?

cmi = **C**omplex **M**odeling **I**nfrastructure

Helpful tools can be accessed through the diffpy.cmi **command line**

cmi -h

Packs

A collection of python packages designed for a specific regression task.

eg. `cmi install pdf`

Profiles

A collection of packs and additional python packages.

eg. `cmi install pdf-plus`

Examples

Example scripts that can be easily copied to your computer.

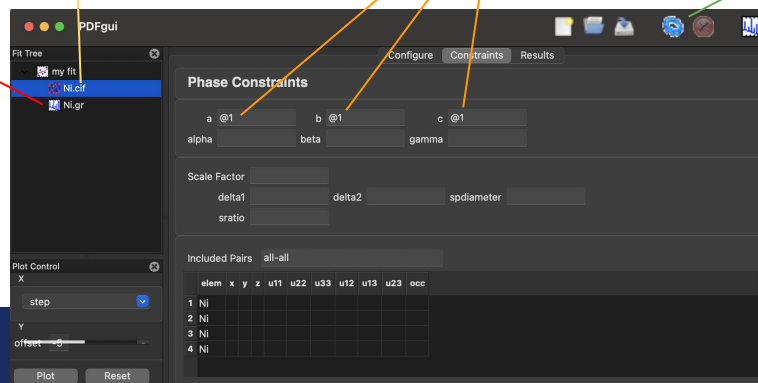
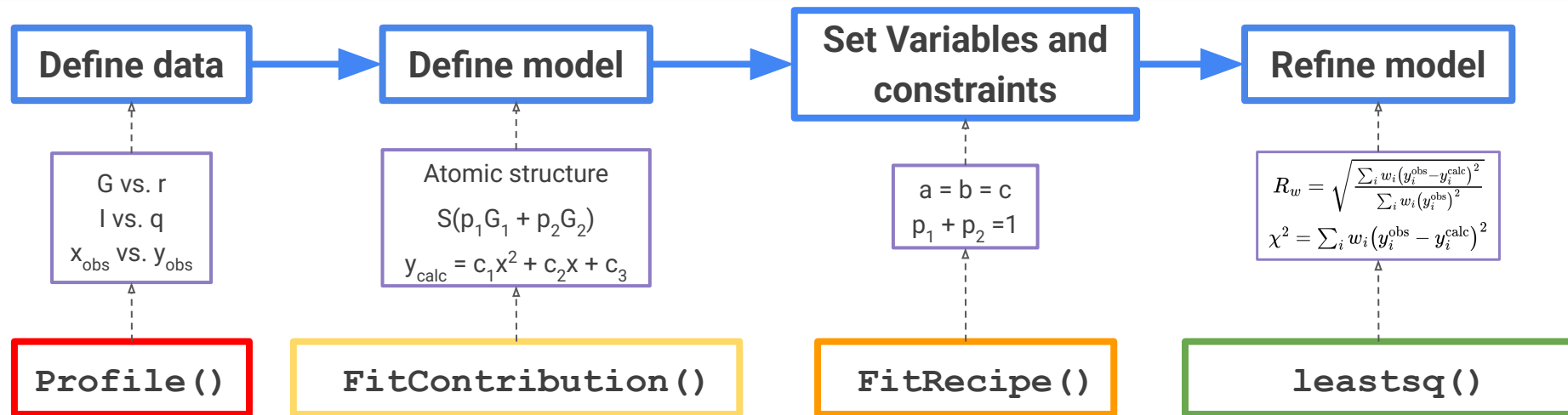
eg. `cmi copy <example-name>`

Info

Prints available packs, profiles, and examples.

eg. `cmi info`

Refinement overview



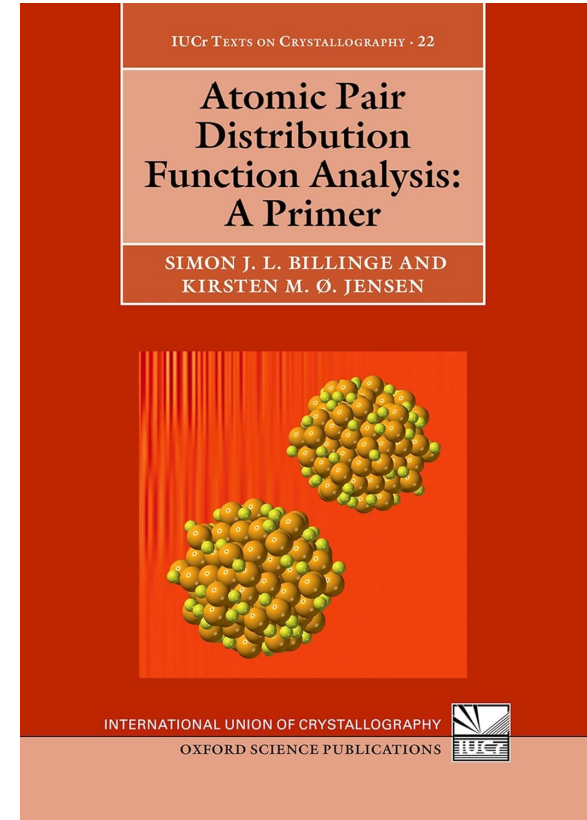
Installing diffpy.cmi

```
conda create -n diffpy.cmi-env python=3.13
conda activate diffpy.cmi-env
conda install -c conda-forge diffpy.cmi
```

Examples!

<https://www.diffpy.org/diffpy.cmi/index.html>

PS: There will likely be a lot of debugging, please
Raise your hand or ask your neighbors for help.



github.com/cadenmyers13/add-2026-diffpycmi